Do Trump's Tweets Spur Market Volatility?

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Abstract

This study examines the daily association between Donald Trump's tweets and stock market volatility from January 2016 to January 2021. We score each tweet with the Valence Aware Dictionary and sEntiment Reasoner (VADER) and assign it a topic with BERTopic. Sessions with a tweet of extreme sentiment, which also carry more tweets, coincide with a rise in the CBOE Volatility Index (VIX). In these sessions, the log change in the VIX is about 0.95% higher than in other sessions, while the coefficients of later sessions are not significant. The timing of the tweets on the coronavirus and on election fraud is consistent with tweets that comment on news rather than create it, as in the literature, although at daily frequency the response of volatility to a tweet cannot be separated from its response to other news in the same session.[1]

1 Introduction

Donald Trump has consistently shared his thoughts on policy, politics and the media through Twitter. Given his global influence, these opinions may offer insight into policy measures that affect the global economy and financial markets, and a growing literature asks whether his tweets move markets. At the market level, Gjerstad et al. (2021) find that the tweets are followed by higher uncertainty, higher trading volume and a decline in the US stock market, with the strongest response to tweets on tariffs and trade. Perico Ortiz (2023) finds that tweets on foreign policy and trade, monetary policy and immigration raise the VIX, with an effect that peaks about 2 hours after posting. Similarly, Nishimura et al. (2021) report that the tweets raise realized volatility and jump risk in the later part of their sample, Burggraf et al. (2020) find that tweets on the trade war predict lower S&P 500 returns and a higher VIX, and Nishimura and Sun (2021) find that tweets on the economic conflict with China raised stock market volatility worldwide.

Whether the tweets carry new information is less clear. Using minute-level data for the stocks that Trump named, Machus et al. (2022) find abnormal returns, higher trading volume and higher investor attention already before the tweets, which is consistent with the tweets commenting on events that had drawn attention. Similarly, Abdi et al. (2023) find that most of his economy-related tweets react to prior market movements rather than convey new information, except for tweets on trade and tariffs, which move the aggregate market.

This study turns to daily frequency, relating the VIX to the sentiment and topics of his tweets from January 2016 to January 2021. We score each tweet with the VADER model of Hutto and Gilbert (2014) and assign it a topic with the BERTopic model of Grootendorst (2022). Event regressions then relate the change in the VIX from 3 sessions before to 3 sessions after the tweet session to an indicator of extreme sentiment and to the number of tweets on each topic. Sessions with an extreme-sentiment tweet show a log change in the VIX 0.95% higher than other sessions, with no significant difference in later sessions. The coronavirus tweets came after the VIX had risen and the election-fraud tweets as it fell, which is consistent with tweets that comment on news rather than create it.

Section 2 describes the dataset. Section 3 scores the sentiment and estimates the event regression. Section 4 extracts the topics and estimates the topic regression. Section 5 concludes.

[1] Code and replication data are available at *GitHub*.

2 Data

Tweets The tweets from @realDonaldTrump are taken from the dataset published on Kaggle (Damarla, 2021), and their times are converted to Eastern Time (ET). Figure 1 shows the tweeting activity over time and the time of posting within the day. Since few tweets in the file precede 2016, the study includes only tweets from 3 January 2016 onward. Within the day, posting peaks in the morning before the open and continues late into the evening. The most frequent words include {*great*, *people*, *america*, *country*, *democrats* and *president*}. Table A.1 in Appendix A describes the full dataset.

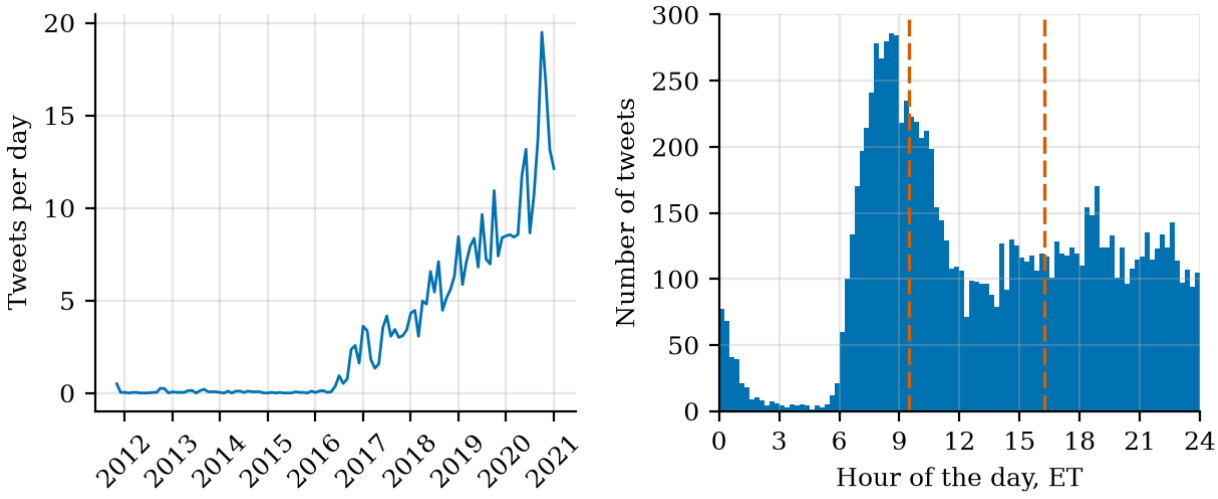


Figure 1: Monthly mean tweets per calendar day (left) and time of posting for tweets from 2016 onward (right), with the open at 9:30 and the VIX close at 16:15 ET marked.

Volatility We proxy market volatility by the VIX, the 30-day implied volatility of the S&P 500. Denote by V_t the close of session t and by r_t its daily log change scaled by 100. Table 1 and Figure 2 describe V_t, r_t and $|r_t|$. The VIX is persistent, spiking in February and December 2018, peaking on 16 March 2020 and remaining elevated through the rest of 2020. Its daily change r_t is right-skewed and heavy-tailed, with a negative first-order autocorrelation, while the autocorrelation of $|r_t|$ is significantly positive over the first 7 lags. This volatility clustering motivates the heteroskedasticity- and autocorrelation-consistent standard errors of Section 3.2.

	Mean	Median	SD	Min	Max	Skewness	Excess kurtosis
V_t	17.70	14.73	8.82	9.14	82.69	2.83	11.92
r_t	0.01	−0.72	8.25	−29.98	76.82	1.51	9.07

Table 1: Summary statistics of the VIX close V_t and its daily log change r_t.

Trading sessions Because the VIX closes at 16:15 ET and many tweets are posted outside trading hours, a tweet cannot simply be matched to its calendar date. We therefore assign each tweet to the first session whose VIX close follows it. With T_i the time of tweet i and $\bar{T}_t$ the VIX

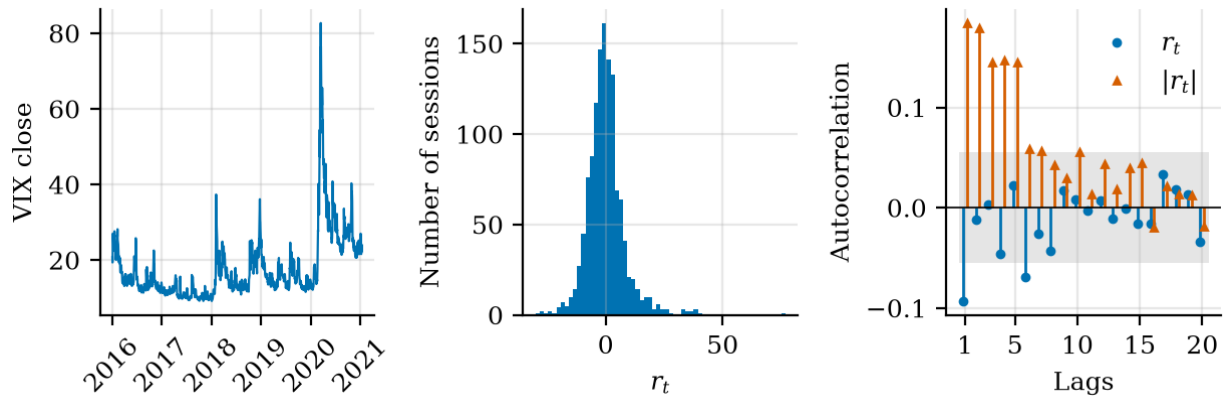


Figure 2: VIX close V_t (left), histogram of the daily log change r_t (middle), and autocorrelations of r_t and $|r_t|$ up to 20 lags, with the 95% band of white noise shaded (right).

close at 16:15 ET on the date of session t, tweet i belongs to session

$$\tau(i) = \min\{t : \bar{T}_t > T_i\}, \qquad \mathcal{T}_t = \{i : \tau(i) = t\}. \tag{1}$$

For example, in a week without holidays, a tweet on a Friday night and a tweet on the following Sunday both belong to the Monday session.

3 Sentiment and Volatility

3.1 Sentiment scores

We score the sentiment of each tweet with VADER (Hutto and Gilbert, 2014), a lexicon of words rated by human judges from -4 to $+4$ and adjusted by rules for capitalization, degree modifiers, negation, contrastive conjunctions and punctuation. Bonta et al. (2019) classify VADER as a "gold standard" lexicon, emphasizing its ability to consider the intensity of words and its validation by humans, and Elbagir and Yang (2019) highlight its effectiveness in classifying the sentiment of tweets on the 2016 US election, noting its ease and speed in processing tweet data.

VADER returns a compound score $s_i \in (-1, 1)$ for each tweet i, and Table 2 summarizes the scores. The compound score labels many tweets as polar, and a threshold of 0.5 already classifies 58% of them as positive or negative. We therefore mark a tweet as extreme only when $|s_i| \geq \theta$ with $\theta = 0.9$, and call a session an event session when it carries at least one such tweet,

$$D_t = \mathbf{1}\{|s_i| \geq \theta \text{ for some } i \in \mathcal{T}_t\}. \tag{2}$$

	Positive	Neutral	Negative
$\theta = 0.5$	3,658	4,389	2,519
$\theta = 0.9$	748	9,414	404

Table 2: Numbers of the 10,566 tweets with $s_i \geq \theta$, $|s_i| < \theta$ and $s_i \leq -\theta$ at two thresholds.

Figure 3 shows the words of the most extreme tweets of each session. Positive tweets center on {*great*, *thank*, *win*, *america* and *people*}, while negative tweets center on {*bad*, *people*, *fake news*, *democrat*, *witch hunt* and *crime*}.



Figure 3: Words from the most positive tweet of each session with $s_i \geq 0.9$ (left) and the most negative tweet of each session with $s_i \leq -0.9$ (right), with size increasing with frequency.

3.2 Event regression

We estimate, for each horizon $k \in \{-3, \ldots, 3\}$, the event regression

$$r_{t+k} = c_k + \phi_k \, r_{t+k-1} + \gamma_k D_t + \varepsilon_{k,t}, \tag{3}$$

where c_k is a constant, ϕ_k the coefficient of the lagged change and $\varepsilon_{k,t}$ an error. The lagged change absorbs the negative autocorrelation of r_t. We estimate the regression by ordinary least squares with the heteroskedasticity- and autocorrelation-consistent standard errors of Newey and West (1987) at 5 lags. Since tweets and volatility may both respond to the news of the day, the coefficient γ_k measures an association rather than a causal effect.

Table 3 reports the results. The coefficient of interest is γ_0, which measures the difference in the log change of the VIX over the tweet session. In an event session, the log change in the VIX is 0.95% higher than in other sessions, which corresponds to 0.14 index points at the median VIX or 1.9% in implied variance. None of the coefficients after the event session is significant, so there is no evidence that a tweet of extreme sentiment predicts the VIX beyond the session that first reflects it. Before the event session, the coefficients alternate in sign and none is significant at the 5% level. Event sessions carry more tweets, 13.5 on average against 4.0 in other sessions, so γ_0 cannot be separated from an association with the number of tweets. News also accumulates while the market is closed, so part of γ_0 may reflect information that arrives over weekends and holidays rather than the content of the tweets.

	$k=-3$	$k=-2$	$k=-1$	$k=0$	$k=1$	$k=2$	$k=3$
ϕ_k	-0.09^{***}	-0.09^{***}	-0.09^{***}	-0.09^{***}	-0.09^{***}	-0.09^{***}	-0.09^{***}
γ_k	-0.23	0.80^{*}	-0.42	0.95^{**}	0.25	0.40	-0.51

Table 3: Event regression results (the constant is not reported), estimated on 1,263 to 1,266 sessions. Significance under the standard errors of Newey and West (1987) at $^{*}p \leq 0.1$, $^{**}p \leq 0.05$ and $^{***}p \leq 0.01$.

Although D_t is associated with the VIX in the tweet session, it does not reveal what the tweets in an event session are about. Event sessions are frequent and cluster in time, as an event session is often followed by another on different subjects and of either sign. Sentiment alone therefore cannot

tell which subjects are associated with volatility, which motivates the topic analysis of the next section.

4 Topics and Volatility

4.1 Topic extraction

To identify the subjects of the tweets, we turn to topic models. Despite its popularity, the latent Dirichlet allocation (LDA) model (Blei et al., 2003) requires the number of topics and the Dirichlet hyperparameters to be set in advance, and its topics may be challenging to interpret, mainly because it relies on a bag-of-words model that ignores word order and assumes that topics are independent. In contrast, BERTopic clusters contextual sentence embeddings, which account for word order, through an architecture of embedding, dimensionality reduction, clustering, tokenization, weighting schemes and fine-tuned representations. Built on pretrained models, deep learning and representation learning, it better captures the semantic nuances and context of the text and identifies more intricate patterns. Egger and Yu (2022) find that BERTopic performs well on Twitter posts, whereas LDA often yields overlapping topics, and Albanese (2022) suggests that BERTopic, along with Top2Vec, performs better with shorter texts such as social media posts or news headlines. We therefore extract the topics with BERTopic.

We implement BERTopic following Grootendorst (2022). The pretrained sentence transformer all-MiniLM-L6-v2 (Reimers and Gurevych, 2019) first embeds each tweet, lowercased and stripped of punctuation, stop words and digits, as a unit vector $\boldsymbol{e}_i \in \mathbb{R}^{384}$. Uniform manifold approximation and projection (UMAP) (McInnes et al., 2018) then maps $\boldsymbol{e}_i$ to $\boldsymbol{u}_i \in \mathbb{R}^3$, and hierarchical density-based clustering (HDBSCAN) (Campello et al., 2013) groups the $\boldsymbol{u}_i$ into clusters of at least 80 tweets, with core distances over neighborhoods of 40 other points. Table 4 lists the resulting 19 topics, $j = 0, \ldots, 18$, each with five words chosen among its 20 words of largest class-based TF-IDF weight. Tweets not associated with any topic, 3,723 of the 10,566, form an outlier class, labeled -1, which is disregarded in the subsequent analysis.

4.2 Topic regression

To measure how each subject relates to volatility, we replace the sentiment indicator with topic counts. With z_i the topic of tweet i, let $N_{j,t}$ be the number of tweets of session t on topic j,

$$N_{j,t} = \big|\{i \in \mathcal{T}_t : z_i = j\}\big|, \qquad j = 0, \ldots, 18. \tag{4}$$

Collecting the counts in $\boldsymbol{X}_t = (N_{0,t}, \ldots, N_{18,t})^\top$, the topic regression is, for each horizon $k \in \{-3, \ldots, 3\}$,

$$r_{t+k} = c_k + \phi_k \, r_{t+k-1} + \boldsymbol{\beta}_k^\top \boldsymbol{X}_t + \varepsilon_{k,t}. \tag{5}$$

where $\boldsymbol{\beta}_k$ collects the coefficients of the 19 topic counts.

Table 5 reports the results. In the tweet session $k = 0$, each additional tweet on the Georgia election, on the border and immigration or on impeachment, holding the other counts fixed, is associated with a log change in the VIX higher by 0.89%, 0.39% and 0.31%, respectively. Before the tweet session, the coefficients significant at the 5% level are all negative, for election fraud, the economy and the Russia investigation. After it, they are positive at $k = 1$ for the Russia investigation and impeachment, and negative at $k = 2$ and $k = 3$ for health care and tax reform, the Middle East, and the border and immigration. The coefficient of -0.34 on the border and immigration at $k = 3$ reverses the positive coefficient of 0.39 in the tweet session rather than extending it.

j	Label	Tweets	Representative words
0	Russia investigation	987	russia, mueller, collusion, fbi, investigation
1	News media	862	fake news, media, cnn, foxnews, nytimes
2	Border and immigration	655	border, wall, immigration, mexico, daca
3	Impeachment	580	impeachment, pelosi, schiff, ukraine, whistleblower
4	Tributes and greetings	510	god, nation, honor, veterans, happy
5	Rallies and endorsements	396	rally, crowd, endorsement, florida, pennsylvania
6	Thanks and short remarks	367	thank, thanks, maga, nice, true
7	Election fraud	351	election, fraud, rigged, ballots, mail-in
8	Economy	334	economy, stock, market, jobs, unemployment
9	Coronavirus	293	coronavirus, testing, cases, vaccine, ventilators
10	Protests and policing	252	anarchists, police, mayor, portland, cuomo
11	China trade	231	china, trade, tariffs, farmers, deal
12	Democratic candidates	228	biden, bloomberg, bernie, debate, sleepy
13	Middle East	217	israel, isis, syria, turkey, kurds
14	North Korea	150	korea, kim, summit, denuclearization, missile
15	Health care and tax reform	141	obamacare, repeal, preexisting, tax, cuts
16	Iran	118	iran, nuclear, deal, sanctions, iraq
17	Supreme Court	87	supreme, court, justices, judges, pack
18	Georgia election	84	georgia, kemp, signature, verification, ballots

Table 4: BERTopic topics with their numbers of tweets and five representative words.

j	Label	$k=-3$	$k=-2$	$k=-1$	$k=0$	$k=1$	$k=2$	$k=3$
	ϕ_k	-0.09^{***}	-0.10^{***}	-0.09^{***}	-0.09^{**}	-0.10^{***}	-0.09^{***}	-0.10^{***}
0	Russia investigation	-0.26^{**}	0.01	-0.22	0.30^{*}	0.25^{**}	0.00	0.22
1	News media	0.09	-0.07	-0.17	-0.31	-0.28	-0.07	0.02
2	Border and immigration	-0.12	0.07	-0.15	0.39^{**}	-0.06	-0.08	-0.34^{**}
3	Impeachment	-0.19	-0.14	0.11	0.31^{**}	0.26^{**}	0.14	-0.07
4	Tributes and greetings	0.08	0.40	-0.46	0.62	-0.56	-0.18	-0.17
5	Rallies and endorsements	0.21	0.72^{*}	0.11	0.36	0.47	-0.33	-0.43
6	Thanks and short remarks	0.35	0.03	-0.16	0.20	-0.53^{*}	0.12	0.17
7	Election fraud	-0.55^{**}	-0.29	-0.43^{**}	-0.33^{*}	-0.25^{*}	0.02	-0.03
8	Economy	-0.18	0.31	-0.73^{**}	-0.03	0.19	0.17	0.27
9	Coronavirus	-0.18	0.53	0.15	0.15	0.61^{*}	0.07	-0.62
10	Protests and policing	-0.17	0.19	-0.22	0.40	-0.13	-0.10	0.44
11	China trade	0.32	-0.34	0.57	0.03	-0.46	-0.44	-0.39
12	Democratic candidates	0.21	-0.08	0.28	-0.47	-0.05	0.56	0.99
13	Middle East	-0.04	-0.28	-0.15	0.14	0.13	-0.68^{**}	-0.89^{**}
14	North Korea	-0.05	0.17	-0.34	0.65	0.25	0.50	0.30
15	Health care and tax reform	-0.50	-0.61	0.67^{*}	0.83	0.45	-1.17^{***}	-0.37
16	Iran	-0.19	-0.44	0.30	-0.46	-0.39	0.07	-0.30
17	Supreme Court	0.32	-0.05	0.23	0.24	0.14	-0.73	-0.31
18	Georgia election	0.26	-0.07	0.66^{*}	0.89^{**}	0.55^{*}	0.06	-0.23

Table 5: Topic regression results (the constant is not reported), estimated on 1,263 to 1,266 sessions. Significance under the standard errors of Newey and West (1987) at $^{*}p \leq 0.1$, $^{**}p \leq 0.05$ and $^{***}p \leq 0.01$.

The timing of the tweets may explain part of these patterns. Some topics are concentrated in time, so their coefficients trace the VIX path over the period in which they were posted. For instance, most tweets on election fraud follow the election of 3 November 2020, after which the VIX fell from 35.6 to 21.6 by 8 January 2021. Their negative coefficients are consistent with a fall

in implied volatility as uncertainty about the outcome resolved while claims of fraud multiplied, suggesting the market did not price the claims as a source of risk.

In contrast, neither the coronavirus nor the China trade topic has a coefficient significant at the 5% level at any horizon. For the coronavirus topic, 77% of the tweets were posted after the VIX peaked on 16 March 2020, so the tweets on the pandemic mostly followed the rise in volatility rather than led it. The China trade topic, to which Gjerstad et al. (2021) and Perico Ortiz (2023) find responses within hours of posting, is not significant at any horizon even at the 10% level. Since the tweet session contains these responses, the absence of an association is consistent with responses that are short-lived or small relative to the daily variation of the VIX.

Overall, the timing of the tweets on the coronavirus and on election fraud points to tweets that react to news rather than create it, in line with Machus et al. (2022) and the "President reacts to news" channel of Abdi et al. (2023).

5 Conclusion

In this study, we examine the daily association between Donald Trump's tweets and the VIX from January 2016 to January 2021, relating the daily log change in the VIX to an indicator of extreme VADER sentiment and to the BERTopic topic counts. Sessions with a tweet of extreme sentiment show a log change in the VIX 0.95% higher than in other sessions, about 0.14 index points at the median VIX, while the coefficients of later sessions are not significant. Among the 19 topics extracted by BERTopic, only a few coefficients are significant, and their signs change across horizons. The tweets on the coronavirus came after the VIX had risen and those on election fraud came as it fell, which is consistent with tweets that comment on news rather than create it.

However, a daily design can hardly separate the response to a tweet from the response to the news it comments on, or the sentiment of the tweets from their number, since tweets and news both accumulate while the market is closed. The market response within hours of posting, documented by Gjerstad et al. (2021) and Perico Ortiz (2023), is also lost at the daily close. Gjerstad et al. (2021) report that an automated trading system based on the tweets achieves positive abnormal returns. A natural extension is therefore to move to intraday E-mini S&P 500 and VIX futures and to test whether a hedging strategy triggered by the tweets pays net of transaction costs.

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A Supplementary Table

Panel A. Tweets in the file	
29 November 2011 to 31 December 2015	80
2016	290
2017	1,076
2018	1,895
2019	2,898
2020	4,310
2021, to 8 January	97
Total	10,646
Panel B. Sample, 3 January 2016 to 8 January 2021	
Tweets	10,566
Tweets per calendar day	5.76
Posted on trading days before the open at 9:30 (%)	23.1
Posted on trading days between the open and the VIX close at 16:15 (%)	23.1
Posted on trading days after the VIX close (%)	24.1
Posted on weekends and holidays (%)	29.8
Panel C. Trading sessions, 4 January 2016 to 14 January 2021	
Sessions	1,268
Sessions with a tweet	1,095
Tweets per session with a tweet, mean	9.65
Tweets per session with a tweet, median	7

Table A.1: The tweet file and the trading sessions (ET throughout). Sessions are defined in (1).

B Latent Dirichlet Allocation

Figure B.1 shows the coherence C_V of Röder et al. (2015) for LDA fitted to the 735 tweets of Figure 3, with $K = 3, \ldots, 18$ topics, each topic being labeled by the three most frequent words among its tweets, on which its coherence is computed. The coherence is highest at $K = 11$, with 0.714, and Table B.1 lists the topics of that model. Six of the 11 topics are labeled by *great*, and 2 topics hold 515 of the 735 tweets. Figure B.2 maps the weights of latent semantic analysis (LSA) (Deerwester et al., 1990) with 7 topics and of LDA with 11 topics by t-distributed stochastic neighbor embedding (t-SNE) (van der Maaten and Hinton, 2008).

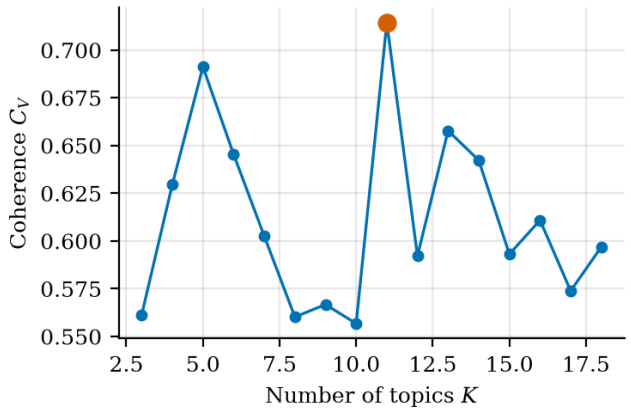


Figure B.1: Coherence C_V of LDA against the number of topics K.

Topic	Tweets	Most frequent words
0	9	great, thank, california
1	18	witch, hunt, people
2	21	great, honor, life
3	163	great, people, america
4	95	america, nation, great
5	10	america, billion, dollar
6	36	national, great, border
7	13	china, death, penalty
8	352	great, america, win
9	13	attack, racist, washington
10	5	vote, illegal, moore

Table B.1: Topics of the LDA model with 11 topics.

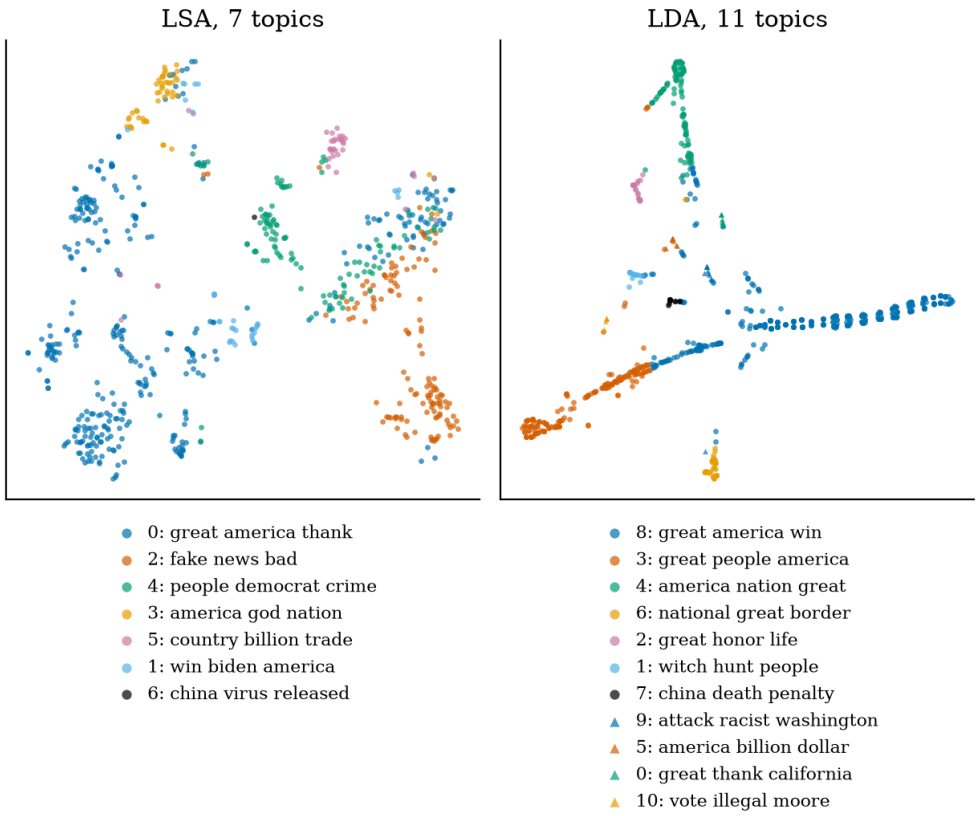


Figure B.2: t-SNE maps of the LSA weights with 7 topics (left) and the LDA weights with 11 topics (right).